

My Project

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Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Person	7
Stud	9

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Person		7
Stud		9

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

functions.cpp	13
header.h	16
OOP-project.cpp	20

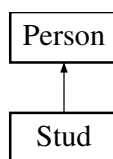
Chapter 4

Class Documentation

4.1 Person Class Reference

```
#include <header.h>
```

Inheritance diagram for Person:



Public Member Functions

- [Person](#) ()
- [Person](#) (string name, string last_name)
- string [get_name](#) () const
- string [get_last_name](#) () const
- virtual void [about](#) () const =0
- virtual [~Person](#) ()

Protected Attributes

- string [name_](#)
- string [last_name_](#)

4.1.1 Constructor & Destructor Documentation

4.1.1.1 [Person\(\)](#) [1/2]

```
Person::Person () [inline]
```

4.1.1.2 Person() [2/2]

```
Person::Person (
    string name,
    string last_name) [inline]
```

4.1.1.3 ~Person()

```
virtual Person::~~Person () [inline], [virtual]
```

4.1.2 Member Function Documentation

4.1.2.1 about()

```
virtual void Person::about () const [pure virtual]
```

Implemented in [Stud](#).

4.1.2.2 get_last_name()

```
string Person::get_last_name () const [inline]
```

4.1.2.3 get_name()

```
string Person::get_name () const [inline]
```

4.1.3 Member Data Documentation

4.1.3.1 last_name_

```
string Person::last_name_ [protected]
```

4.1.3.2 name_

```
string Person::name_ [protected]
```

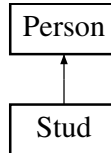
The documentation for this class was generated from the following file:

- [header.h](#)

4.2 Stud Class Reference

```
#include <header.h>
```

Inheritance diagram for Stud:



Public Member Functions

- [Stud](#) ()
- [Stud](#) (string name, string last_name)
- [Stud](#) (std::stringstream &is, int number_of_homework_marks)
- [Stud](#) (const [Stud](#) &other)
- [Stud](#) ([Stud](#) &&other) noexcept
- [Stud](#) & [operator=](#) (const [Stud](#) &other)
- [Stud](#) & [operator=](#) ([Stud](#) &&other) noexcept
- int [get_exam_mark](#) () const
- void [set_exam_mark](#) (int exam_mark)
- void [set_homework_marks](#) (int Homework_mark)
- void [clean_homework_marks](#) ()
- void [set_final_mark](#) (double final_mark)
- double [get_final_mark](#) () const
- void [set_second_final_mark](#) (double second_final_mark)
- double [get_second_final_mark](#) () const
- double [Get_average_for_homework_mark](#) ()
- double [Get_mediana_for_homework_mark](#) ()
- void [generate_marks](#) ()
- void [generate_name](#) ()
- void [about](#) () const override
- [~Stud](#) ()

Public Member Functions inherited from [Person](#)

- [Person](#) ()
- [Person](#) (string name, string last_name)
- string [get_name](#) () const
- string [get_last_name](#) () const
- virtual [~Person](#) ()

Friends

- std::ostream & [operator<<](#) (std::ostream &out, const [Stud](#) &student)
- std::istream & [operator>>](#) (std::istream &in, [Stud](#) &student)

Additional Inherited Members

Protected Attributes inherited from [Person](#)

- string [name_](#)
- string [last_name_](#)

4.2.1 Constructor & Destructor Documentation

4.2.1.1 Stud() [1/5]

```
Stud::Stud () [inline]
```

4.2.1.2 Stud() [2/5]

```
Stud::Stud (  
    string name,  
    string last_name) [inline]
```

4.2.1.3 Stud() [3/5]

```
Stud::Stud (  
    std::stringstream & is,  
    int number_of_homework_marks)
```

4.2.1.4 Stud() [4/5]

```
Stud::Stud (  
    const Stud & other) [inline]
```

4.2.1.5 Stud() [5/5]

```
Stud::Stud (  
    Stud && other) [inline], [noexcept]
```

4.2.1.6 ~Stud()

```
Stud::~Stud ()
```

4.2.2 Member Function Documentation

4.2.2.1 about()

```
void Stud::about () const [inline], [override], [virtual]
```

Implements [Person](#).

4.2.2.2 clean_homework_marks()

```
void Stud::clean_homework_marks () [inline]
```

4.2.2.3 generate_marks()

```
void Stud::generate_marks ()
```

4.2.2.4 generate_name()

```
void Stud::generate_name ()
```

4.2.2.5 Get_average_for_homework_mark()

```
double Stud::Get_average_for_homework_mark ()
```

4.2.2.6 get_exam_mark()

```
int Stud::get_exam_mark () const [inline]
```

4.2.2.7 get_final_mark()

```
double Stud::get_final_mark () const [inline]
```

4.2.2.8 Get_mediana_for_homework_mark()

```
double Stud::Get_mediana_for_homework_mark ()
```

4.2.2.9 get_second_final_mark()

```
double Stud::get_second_final_mark () const [inline]
```

4.2.2.10 operator=() [1/2]

```
Stud & Stud::operator= (  
    const Stud & other)
```

4.2.2.11 operator=() [2/2]

```
Stud & Stud::operator= (  
    Stud && other) [noexcept]
```

4.2.2.12 set_exam_mark()

```
void Stud::set_exam_mark (
    int exam_mark) [inline]
```

4.2.2.13 set_final_mark()

```
void Stud::set_final_mark (
    double final_mark) [inline]
```

4.2.2.14 set_homework_marks()

```
void Stud::set_homework_marks (
    int Homework_mark) [inline]
```

4.2.2.15 set_second_final_mark()

```
void Stud::set_second_final_mark (
    double second_final_mark) [inline]
```

4.2.3 Friends And Related Symbol Documentation

4.2.3.1 operator<<

```
std::ostream & operator<< (
    std::ostream & out,
    const Stud & student) [friend]
```

4.2.3.2 operator>>

```
std::istream & operator>> (
    std::istream & in,
    Stud & student) [friend]
```

The documentation for this class was generated from the following files:

- [header.h](#)
- [functions.cpp](#)

Chapter 5

File Documentation

5.1 functions.cpp File Reference

```
#include "header.h"
```

Functions

- void [Get_final_mark](#) (vector< [Stud](#) > &grupe, bool for_average_homework_mark, bool for_both_homework_mark)
- void [Print_final_mark](#) (vector< [Stud](#) > &grupe, bool for_average_homework_mark, bool for_both_homework_mark, bool print_results_in_terminal)
- int [Get_size_for_string_printing](#) (vector< [Stud](#) > &grupe)
- void [generate_marks](#) (vector< int > &Marks, int number_of_marks)
- void [Sort_students](#) (vector< [Stud](#) > &grupe, string parametr)
- bool [Generate_file_with_students](#) (int number_of_students, int number_of_marks, string filename)
- void [Divide_for_two_grupse](#) (vector< [Stud](#) > &grupe, vector< [Stud](#) > &best_grupe, vector< [Stud](#) > &worst_grupe)
- void [Divide_for_two_grupse](#) (vector< [Stud](#) > &grupe, vector< [Stud](#) > &worst_grupe)
- void [Divide_for_two_grupse_v3](#) (vector< [Stud](#) > &grupe, vector< [Stud](#) > &best_grupe, vector< [Stud](#) > &worst_grupe)
- void [Enter_students_using_txt_file_bufer_P](#) (vector< [Stud](#) > &grupe)
- std::ostream & [operator<<](#) (std::ostream &out, const [Stud](#) &student)
- std::istream & [operator>>](#) (std::istream &in, [Stud](#) &student)

Variables

- double [time_of_generating_file](#) = 0
- double [time_of_reading_file](#) = 0
- double [time_of_dividing](#) = 0
- double [time_of_sorting](#) = 0
- double [time_of_culculating](#) = 0
- double [time_of_writing_files](#) = 0

5.1.1 Function Documentation

5.1.1.1 Divide_for_two_grupse() [1/2]

```
void Divide_for_two_grupse (  
    vector< Stud > & grupe,  
    vector< Stud > & best_grupe,  
    vector< Stud > & worst_grupe)
```

5.1.1.2 Divide_for_two_grupse() [2/2]

```
void Divide_for_two_grupse (  
    vector< Stud > & grupe,  
    vector< Stud > & worst_grupe)
```

5.1.1.3 Divide_for_two_grupse_v3()

```
void Divide_for_two_grupse_v3 (  
    vector< Stud > & grupe,  
    vector< Stud > & best_grupe,  
    vector< Stud > & worst_grupe)
```

5.1.1.4 Enter_students_using_txt_file_bufer_P()

```
void Enter_students_using_txt_file_bufer_P (  
    vector< Stud > & grupe)
```

5.1.1.5 Generate_file_with_students()

```
bool Generate_file_with_students (  
    int number_of_students,  
    int number_of_marks,  
    string filename)
```

5.1.1.6 generate_marks()

```
void generate_marks (  
    vector< int > & Marks,  
    int number_of_marks)
```

5.1.1.7 Get_final_mark()

```
void Get_final_mark (  
    vector< Stud > & grupe,  
    bool for_average_homework_mark,  
    bool for_both_homework_mark)
```

5.1.1.8 Get_size_for_string_printing()

```
int Get_size_for_string_printing (
    vector< Stud > & grupe)
```

5.1.1.9 operator<<()

```
std::ostream & operator<< (
    std::ostream & out,
    const Stud & student)
```

5.1.1.10 operator>>()

```
std::istream & operator>> (
    std::istream & in,
    Stud & student)
```

5.1.1.11 Print_final_mark()

```
void Print_final_mark (
    vector< Stud > & grupe,
    bool for_average_homework_mark,
    bool for_both_homework_mark,
    bool print_results_in_terminal)
```

5.1.1.12 Sort_students()

```
void Sort_students (
    vector< Stud > & grupe,
    string parametr)
```

5.1.2 Variable Documentation

5.1.2.1 time_of_calculating

```
double time_of_calculating = 0
```

5.1.2.2 time_of_dividing

```
double time_of_dividing = 0
```

5.1.2.3 time_of_generating_file

```
double time_of_generating_file = 0
```

5.1.2.4 time_of_reading_file

```
double time_of_reading_file = 0
```

5.1.2.5 time_of_sorting

```
double time_of_sorting = 0
```

5.1.2.6 time_of_writing_files

```
double time_of_writing_files = 0
```

5.2 header.h File Reference

```
#include <iostream>
#include <iomanip>
#include <string>
#include <vector>
#include <random>
#include <fstream>
#include <sstream>
#include <chrono>
#include <algorithm>
```

Classes

- class [Person](#)
- class [Stud](#)

Functions

- void [Get_final_mark](#) (vector< [Stud](#) > &grupe, bool for_average_homework_mark, bool for_both_↵ homework_mark)
- void [Print_final_mark](#) (vector< [Stud](#) > &grupe, bool for_average_homework_mark, bool for_both_↵ homework_mark, bool print_results_in_terminal)
- int [Get_size_for_string_printing](#) (vector< [Stud](#) > &grupe)
- void [generate_marks](#) (vector< int > &Marks, int number_of_marks)
- void [Sort_students](#) (vector< [Stud](#) > &grupe, string parametr)
- bool [Generate_file_with_students](#) (int number_of_students, int number_of_marks, string filename)
- void [Divide_for_two_grupse](#) (vector< [Stud](#) > &grupe, vector< [Stud](#) > &best_grupe, vector< [Stud](#) > &worst_grupe)
- void [Divide_for_two_grupse](#) (vector< [Stud](#) > &grupe, vector< [Stud](#) > &worst_grupe)
- void [Divide_for_two_grupse_v3](#) (vector< [Stud](#) > &grupe, vector< [Stud](#) > &best_grupe, vector< [Stud](#) > &worst_grupe)
- void [Enter_students_using_txt_file_bufer_P](#) (vector< [Stud](#) > &grupe)

Variables

- double `time_of_generating_file`
- double `time_of_reading_file`
- double `time_of_dividing`
- double `time_of_sorting`
- double `time_of_culculating`
- double `time_of_writing_files`

5.2.1 Function Documentation

5.2.1.1 `Divide_for_two_grupse()` [1/2]

```
void Divide_for_two_grupse (  
    vector< Stud > & grupe,  
    vector< Stud > & best_grupe,  
    vector< Stud > & worst_grupe)
```

5.2.1.2 `Divide_for_two_grupse()` [2/2]

```
void Divide_for_two_grupse (  
    vector< Stud > & grupe,  
    vector< Stud > & worst_grupe)
```

5.2.1.3 `Divide_for_two_grupse_v3()`

```
void Divide_for_two_grupse_v3 (  
    vector< Stud > & grupe,  
    vector< Stud > & best_grupe,  
    vector< Stud > & worst_grupe)
```

5.2.1.4 `Enter_students_using_txt_file_bufer_P()`

```
void Enter_students_using_txt_file_bufer_P (  
    vector< Stud > & grupe)
```

5.2.1.5 `Generate_file_with_students()`

```
bool Generate_file_with_students (  
    int number_of_students,  
    int number_of_marks,  
    string filename)
```

5.2.1.6 `generate_marks()`

```
void generate_marks (  
    vector< int > & Marks,  
    int number_of_marks)
```

5.2.1.7 Get_final_mark()

```
void Get_final_mark (
    vector< Stud > & grupe,
    bool for_average_homework_mark,
    bool for_both_homework_mark)
```

5.2.1.8 Get_size_for_string_printing()

```
int Get_size_for_string_printing (
    vector< Stud > & grupe)
```

5.2.1.9 Print_final_mark()

```
void Print_final_mark (
    vector< Stud > & grupe,
    bool for_average_homework_mark,
    bool for_both_homework_mark,
    bool print_results_in_terminal)
```

5.2.1.10 Sort_students()

```
void Sort_students (
    vector< Stud > & grupe,
    string parametr)
```

5.2.2 Variable Documentation

5.2.2.1 time_of_culculating

```
double time_of_culculating [extern]
```

5.2.2.2 time_of_dividing

```
double time_of_dividing [extern]
```

5.2.2.3 time_of_generating_file

```
double time_of_generating_file [extern]
```

5.2.2.4 time_of_reading_file

```
double time_of_reading_file [extern]
```


5.2.2.5 time_of_sorting

```
double time_of_sorting [extern]
```

5.2.2.6 time_of_writing_files

```
double time_of_writing_files [extern]
```

5.3 header.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <iostream>
00004 #include <iomanip>
00005 #include <string>
00006 #include <vector>
00007 #include <random>
00008 #include <fstream>
00009 #include <sstream>
00010 #include <chrono>
00011 #include <algorithm>
00012
00013 using std::cin;
00014 using std::cout;
00015 using std::endl;
00016 using std::exception;
00017 using std::fixed;
00018 using std::left;
00019 using std::move;
00020 using std::setprecision;
00021 using std::setw;
00022 using std::string;
00023 using std::vector;
00024
00025 class Person
00026 {
00027 protected:
00028     string name_;
00029     string last_name_;
00030
00031 public:
00032     Person() {};
00033     Person(string name, string last_name) : name_(name), last_name_(last_name) {};
00034     inline string get_name() const { return name_; };
00035     inline string get_last_name() const { return last_name_; };
00036     virtual void about() const = 0;
00037     virtual ~Person() {};
00038 };
00039
00040 class Stud : public Person
00041 {
00042 private:
00043     vector<int> Homework_marks_;
00044     int exam_mark_;
00045     double final_mark_ = 0;
00046     double second_final_mark_ = -1;
00047
00048 public:
00049     Stud() {};
00050     Stud(string name, string last_name) : Person(name, last_name), exam_mark_(0) {};
00051     Stud(std::stringstream &is, int number_of_homework_marks);
00052     Stud(const Stud &other) : Person(other.name_, other.last_name_),
00053                               Homework_marks_(other.Homework_marks_), exam_mark_(other.examen_mark_),
00054                               final_mark_(other.final_mark_), second_final_mark_(other.second_final_mark_) {};
00055     Stud(Stud &&other) noexcept : Person(move(other.name_), move(other.last_name_)),
00056                                   Homework_marks_(move(other.Homework_marks_)),
00057                                   exam_mark_(other.examen_mark_), final_mark_(other.final_mark_),
00058                                   second_final_mark_(other.second_final_mark_)
00059     {
00060         other.examen_mark_ = 0;
00061         other.final_mark_ = 0;
00062         other.second_final_mark_ = 0;
00063     };
00064 }
```

```

00061     Stud &operator=(const Stud &other);
00062     Stud &operator=(Stud &&other) noexcept;
00063     friend std::ostream &operator<<(std::ostream &out, const Stud &student);
00064     friend std::istream &operator>>(std::istream &in, Stud &student);
00065     inline int get_exam_mark() const { return exam_mark_; };
00066     inline void set_exam_mark(int exam_mark) { exam_mark_ = exam_mark; };
00067     inline void set_homework_marks(int Homework_mark) { Homework_marks_.push_back(Homework_mark); };
00068     inline void clean_homework_marks() { Homework_marks_.clear(); };
00069     inline void set_final_mark(double final_mark) { final_mark_ = final_mark; };
00070     inline double get_final_mark() const { return final_mark_; };
00071     inline void set_second_final_mark(double second_final_mark) { second_final_mark_ =
second_final_mark; };
00072     inline double get_second_final_mark() const { return second_final_mark_; };
00073     double Get_average_for_homework_mark();
00074     double Get_mediana_for_homework_mark();
00075     void generate_marks();
00076     void generate_name();
00077     inline void about() const override { cout << "I am a student" << endl; };
00078     ~Stud();
00079 };
00080
00081 extern double time_of_generating_file;
00082 extern double time_of_reading_file;
00083 extern double time_of_dividing; // For best and worst
00084 extern double time_of_sorting;
00085 extern double time_of_culculating;
00086 extern double time_of_writing_files;
00087
00088 void Get_final_mark(vector<Stud> &grupe, bool for_average_homework_mark, bool for_both_homework_mark);
00089 void Print_final_mark(vector<Stud> &grupe, bool for_average_homework_mark, bool
for_both_homework_mark, bool print_results_in_terminal);
00090 int Get_size_for_string_printing(vector<Stud> &grupe);
00091 void generate_marks(vector<int> &Marks, int number_of_marks);
00092 void Sort_students(vector<Stud> &grupe, string parametr);
00093 bool Generate_file_with_students(int number_of_students, int number_of_marks, string filename);
00094 void Divide_for_two_grupse(vector<Stud> &grupe, vector<Stud> &best_grupe, vector<Stud> &worst_grupe);
00095 void Divide_for_two_grupse(vector<Stud> &grupe, vector<Stud> &worst_grupe);
00096 void Divide_for_two_grupse_v3(vector<Stud> &grupe, vector<Stud> &best_grupe, vector<Stud>
&worst_grupe);
00097 void Enter_students_using_txt_file_bufer_P(vector<Stud> &grupe);

```

5.4 OOP-project.cpp File Reference

```
#include "header.h"
```

Functions

- int [main](#) ()

5.4.1 Function Documentation

5.4.1.1 main()

```
int main ()
```

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